

- 4 (a) Consider the top most position.

When the angular speed is just enough for person to stay in contact with the wall, person is about to lose contact with the wall.

As such, normal contact force = 0 at top most position

Component of weight parallel to floor of cylinder provides for the centripetal force [B1]

$$mg \sin \theta = mr\omega^2$$

$$(9.81) \sin 50^\circ = (8.0) \omega^2 \quad [\text{M1}]$$

$$\omega = 0.97 \text{ rad s}^{-1} \quad [\text{A0}]$$

- (b) At the bottom most position,

The resultant of the normal contact force from wall and component of weight parallel to floor provides for the centripetal force.

$$N_{\text{wall}} - mg \sin \theta = mr\omega^2 \quad [\text{C1}]$$

$$N_{\text{wall}} = (65)(8.0)(0.97)^2 + (65)(9.81) \sin 50^\circ$$

$$N_{\text{wall}} = 980 \text{ N} \quad [\text{A1}]$$

- (c) Gain in gravitational potential energy

$$= 65(9.81)(2 \times 8.0 \times \sin 50^\circ) \quad [\text{C1}]$$

$$= 7800 \text{ J} \quad [\text{A1}]$$

- (d) Average power

$$= \frac{\text{work done}}{\text{time}}$$

$$= \frac{\text{gain in gravitational potential energy}}{\text{time}}$$

$$= \frac{7800}{\frac{1}{2} \left(\frac{2\pi}{0.97} \right)} \quad [\text{C1}]$$

$$= 2400 \text{ W} \quad [\text{A1}]$$

Allow ecf from (c)

